

Master AAI Expert Workshops — Session 1

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Introduction Tech Workshop Curriculum

Curriculum Overview

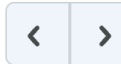
No.	Date	Lecturer	Subjects
1	Thu Sep	Rick + Michiel	AI assisted coding, Data exploration, Faces
2	Thu Sep	Michiel	Object Detection, Video, CNN advanced
3	Thu Sep	Rick	Evaluations, Build an app
4	Thu Sep	Michiel	Embeddings (SSL, CLIP, Search) & prep tech reviews
5	Thu Oct	Rick + Michiel	Tech reviews
6	Thu Oct	Rick	ML Ops + topics based on student questions
7	Thu Oct	Michiel	Vision Language Models (VLMs) + topics based on student questions

Vragenbox op DLO

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Anonieme vragenbox ▾



wooclap



Vragenlijsten

Anonieme vragenbox

Bezig: Vraag 0/1



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Techreview week 5

Assessment on the technical aspect of your project up till that point. Outcomes A2, B1, B2, B3, C2

You hand in on DLO:

- Relevant part of your project report (see study guide)
- Link to your up-to-date Git repo (make it navigable)

Usage of Generative AI

AI Assessment Scale Level 3: AI mag gebruikt worden ter ondersteuning bij specifieke taken. Je wordt altijd verplicht om AI-gegenereerde content kritisch te evalueren en indien nodig aan te passen (Zie Techworkshop DLO).

Verantwoording van keuzes in eigen woorden om begrip aan te kunnen tonen — Gen AI niet toegestaan voor schrijven uiteindelijke uitleg in rapport.

AI Assisted Coding

AI Assisted Coding in VS Code

The screenshot displays the Visual Studio Code interface with a Python file named 'Untitled-1'. The code defines a `fibonacci(n)` function that calculates the nth Fibonacci number. The function includes docstrings for its purpose, arguments, and return values. It uses a loop to calculate the sequence iteratively. A main block prompts the user to enter a number and prints the result.

```
def fibonacci(n):
    """
    Calculate the nth Fibonacci number.

    Args:
        n (int): The index of the Fibonacci number to calculate.

    Returns:
        int: The nth Fibonacci number.
    """
    if n <= 0:
        return 0
    elif n == 1:
        return 1
    else:
        a, b = 0, 1
        for _ in range(2, n + 1):
            a, b = b, a + b
        return b

if __name__ == "__main__":
    num = int(input("Enter a number: "))
    print(f"The {num}th Fibonacci number is {fibonacci(num)}")
```

On the right, the 'CHAT' panel shows a conversation with 'MichielBontenbal'. The user has asked for a Python file to calculate Fibonacci numbers. The AI, 'GitHub Copilot', has provided the code shown in the editor. Below the code, it asks 'How can I optimize this code?' and provides options to 'Add Context...' or 'def fibonacci(n): Current file'.

At the bottom of the chat panel, there is a text input field with the placeholder 'Ask Copilot', a microphone icon, and buttons for 'Ask', 'Gemini 2.0 Flash', and a send button.

Try a different model!

Do you get the right response? ‘Alignment’



Intro to Alignment: aligned.substack.com — blog by Jan Leike

Tips for Better Prompting

Write a better prompt

- Think before you prompt
- Try different prompts
- For beginners: add “write simple code...”

Understand the response

- Be aware of mistakes
- Ask “Explain this code to me”
- Decide if you want to use the code or try again

Human–AI Pair Programming



Work together with the AI — your new best buddy!

Active Learning is Better!

Writing code is more effective than reading code. Why?

- You remember it better
- You are more critical
- You are more engaged

First Learn to Walk Before You Learn to Run



To Do / Exercises

Beginners

- Set up Copilot in VS Code
- Write code to calculate Fibonacci numbers

Experienced

- Compare: Claude.ai vs Copilot
- Try uploading an image to generate code
- Create a front-end app with HTML/JS/CSS (multiple files)

Data Exploration and Initial Analysis

The Life-Cycle of MLOps: CRISP-ML(Q)

CRISP-ML(Q)	CRISP-DM	Amershi <i>et al.</i> [28]	Breck <i>et al.</i> [29]	
Business & Data Understanding	Business Un- derstanding	Require- ments	-	
	Data Under- standing	Collection	Data	Infra- structure
Data Preparation	Data Preparation	Cleaning		
		Labeling		
		Feature Engi- neering		
Modeling	Modeling	Training	Model	
Evaluation	Evaluation	Evaluation	-	
Deployment	Deployment	Deployment	-	
Monitoring & Maintenance	-	Monitoring	Monitoring	

Choosing a Dataset

- Always first concern yourself with **Data Provenance**
 - Source, collection process — might any of these be biased?
- Then check **Data Description**
 - Size, contents, label quality
- **Where to find data?**
 - Similar research, Kaggle, Hugging Face

Initial Exploration and Visualisation

- Visualisation is an integral part of the data exploration phase
- First look at different distributions: feature distributions, bivariate/multivariate exploration
- Use histograms, boxplots, scatterplots, etc.

Also check technical details:

- Target leaking
- Image quality (resolution, brightness, cropping)
- Label noise (incorrect ages, approximations)

Ask yourself: what makes this data predictive?

Data Bias When Using People

- Algorithms do not care about your data at all
- They just want to “win” by solving your problem
- Winning = minimising your chosen loss function
- By finding any pattern that might contribute to this
- **Representation is therefore unimaginably important**

Feature Engineering

When you know your data, you can start engineering:

- Face alignment / Cropping
- Grayscale
- Over- and Undersampling of classes
- Normalisation of the data
- Regression vs Classification framing

Don't just ask yourself what — but more importantly why.

Other Important Subjects

- Data leakage / test data invalidation
- Data splitting strategies (random, stratified, etc.)
- Think already about evaluation metrics (MAE, RMSE, Fairness metrics)
- Reproducibility of your workflow
- The iterative nature of it (learning outcome B4)

Practicum ~45 min

Make groups of four students. Perform data analysis on the **UTKFace dataset** (see notebook on DLO).

Questions to answer:

- **Distribution:** What distributions should be checked? Balanced or skewed?
- **Quality:** Any errors, anomalies, missing data, unrealistic images?
- **Bias:** What kind of biases are present? How would you fix them?
- **Image inspection:** Lighting, angles, cropping, etc.

In the end we recap: surprises? concerns? take-aways for the project?

Faces

Faces — Time Lapse



Python Packages for Faces

No	Name	Description	Link
1	MTCNN	CNN model for face detection	pypi.org/project/mtcnn
2	Deepface	CNN model for age estimation	pypi.org/project/deepface
3	face-recognition	Face recognition library	pypi.org/project/face-recognition

Exercise Notebooks

- `Gezichtsdetectie_with_cnn.ipynb`
- `Age_estimation_Deepface.ipynb`